

ENKO Ltd

Projectname:

DEIF NEW BATTERY CHARGERS

Document name:

TECHNICAL SPECIFICATIONS AND MODELS DESCRIPTION

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EXPLANATIONS:

THIS DOCUMENT PRESENTS THE FINAL TECHNICAL SPECIFICATIONS FOR THE NEW BATTERY CHARGER SERIES TO BE DESIGNED FOR DEIF. DOCUMENT CONTAINS ELECTRICAL AND MECHANICAL SPECIFICATIONS.

THIS DOCUMENT IS REVISED TO "REV01" ON 18/AUG/2021

DOCUMENT IS REVISED BASED ON MAIL SENT BY LENE SKJERNAA ON 20/08/2021
REV-04 EXPLAINS THE EXPECTED COST IMPACT AND DIFFERENCES IN TECHNICAL SPECIFICATIONS WITH REFERENCE TO PPT PRESENTATION DOCUMENT PREVIOUSLY SHARED WITH DEIF

Revision:

R00
R01
R02
R03
R04

Preparation:

ENKO R&D

Contents:

- PROJECT SCOPE
- MODEL DESCRIPTIONS
- TECHNICAL SPECIFICATIONS (REVISED TO "REV-04")
- COMMENTS

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PROJECT BACKGROUND:

This project was initiated in order to create a new `Battery Charger` series for DEIF with up-to-date technology and specifications and at the same time, renew the mechanical construction design to suit market demand.

Both DEIF and ENKO technical teams participated in a series of discussions to determine the final technical specifications for these new charger series and final specifications document is generated and published in the following chapters. A revision can be made to these specifications before the project starts and all changes will be followed by revision updates, based on this document.

GENERAL DESCRIPTION OF THE MODELS:

Based on the technical discussions till date, the model arrangement of the new battery chargers is given in the table below:

MODELS:	75W	150W	300W	600W	1200W
MODEL – A	12V/5A				
MODEL – B		12V10A 24V/5A			
MODEL – C			24V/10		
MODEL – D				24V/20A	
MODEL – E					24V/40A

The above model names will be nominated by DEIF in the later stages of the project. MODEL-B is a 150W package device and charge voltage will be user selectable in the field.

NOTES ON CHARGER SPECIFICATIONS:

Rev-03 contains all the revisions we have been discussing during our meetings with DEIF project team. This is the final technical specifications document for DEIF Charger project.

Model names are given as reference to this document and DEIF should determine each model name and description.

Mechanical dimensions are given as reference in this document and only approximate dimensions are given. During design phase, exact dimensions will be calculated and published in respective documents. Charger enclosure design work will be made in ENKO and plastic injection molds will be manufactured in ENKO.

Reinforced insulation term is used instead of "Double Insulation" as double insulation refers to non-earthed devices where output is completely FLOATING, and input is not connected to EARTH point. This is not the case in DEIF charger series.

Revision Rev-04 includes notes on added technical specifications with reference to the initial specifications document (ppt document) shared with DEIF initially and the estimated cost impact on the models, based on normal component prices (impact of component shortage premium costs is not included in this calculation).

MODEL-A: 12V/5A (75W PACKAGE)

This is the basic model and will be based on flyback converter topology. Targeted efficiency will be better than 86% at 220Vac input voltage and above 50% load conditions.

Specification	Description		
<u>Input Specifications:</u>			
AC Supply Input Voltage range:	Universal AC power input (withstand to 300Vac)	88.0Vac min.	265Vac max.
		@ full-load conditions	
Operating Input Frequency range:	50Hz / 60Hz	45Hz	70Hz
DC Input voltage range:	DC volts	120V	400V
		@ full-load conditions	
Continuous Input current at full load condition:	@ 5A load condition	@ 90Vac input	<1.8Aac (typ.)
		@ 220Vac input	<0.7Aac (typ.)
Efficiency:	>83% worst case condition (typical efficiency: 85%)		
Input Power Factor:	>0,65 Capacitive (no PFC compensation)		
Input fuse protection:	PCB mounted, non-replaceable fuse (no user serviceable parts)		
Rated input power:	<88W max. at full load. (@ worst case efficiency)		
Input terminal type:	Enclosure front panel mounted	Mains input	GSTB 2.5 / 3 (12Adc) Phoenix three pole screw type terminal set
		DC output	MSTB 2.5 / 6 (12Adc) Phoenix six pole screw type terminal set
<u>Output Specifications:</u>			
Output voltage range:	12V model only	Factory set at 13.8Vdc float charge Boost mode: 14.5Vdc (gassing level)	
Output current range:		0Adc to 5Adc (load dependent)	
Short-circuit current:	100% rated output current into short circuit, continuous operation.		
Output ripple voltage:	<1.0% of nominal output voltage over 10Hz to 100KHz		
Output voltage tolerance:	<±1.5% of set value		

Current limit:	100% rated current during BULK and ABSORPTION charging modes	
Boost mode control:	Built-in automatic boost function, (boost mode charge current derating value will be decided during design phase.)	
Output noise level:	<1.2V pk-pk (30KHz to 300KHz) connected to resistive load	
Line regulation:	<1% over 100Vac to 265Vac input voltage range at constant output load (not exceeding rated load)	
Load regulation:	<1% at constant input voltage, from 10% load to 90% load on the output, measured at the output terminals	
Reverse polarity protection:	Active protection	"Automatic Resettable" SS reverse polarity protection. No external component (fuse) is required.
Front panel indicators:	Multicolor LED indicator for status and alarm conditions: • Float mode (normal operation): steady green • Boost mode: blinking green • Failure: blinking red	
Auxiliary output:	Alarm output, N/O contact relay (SPST), 250Vac@1A, AC3 class	
<u>Environmental conditions:</u>		
Operating temperature range:	-40°C to +70°C with output power derating	
Output Power derating:	Output power is linearly de-rated starting at 55°C down to 50% rated current at T _{AMB} = 70°C (approx. -3.5%/°C power derating above 55°C))	
Storage temperature and humidity:	-40°C to +85°C, (0%Rh to 95%Rh, non-condensing)	
Cooling type:	Convection cooled (vertical mounted, no physical obstacles are allowed within 20mm from both sides and minimum 50mm for top and bottom sides)	
Operating humidity range:	0%Rh to 95%Rh, non-condensing with AC supply connected. Derating may activate below 30%RH depending on operating conditions.	
Altitude:	<2000mt at full rated load. Derate linearly from full power to 50% power from 2000mt to 3000mt over the specified working temperature range.	
Dielectric withstanding voltage:	Mains input – DC output, Boost input & Alarm o/p (with voltage input terminals shorted) Reinforced-insulation between primary and secondary circuits.	4.0 KV _{DC} @ 70% RH
	Mains input – Earth (voltage input terminals shorted)	2,5 KV _{DC} @ 70% RH
	DC Output – Earth (with output terminals shorted)	0,5 KV _{AC} @ 70% RH
	Alarm o/p - DC out & Boost terminals	2,5 KV _{DC} @ 70% RH
ESD protection:	8kV free air discharge onto the metal case and plastic housing (metal case connected to EARTH)	
EMI noise immunity:	IEC/EN 61000-6-2,	
Conducted and radiated EMC:	IEC/EN61000-6-4, IEC/EN61000-4-4, IEC/EN61000-4-5, IEC/EN61000-4-6, IEC/EN61000-4-11, IEC/EN61000-3-2, Light industry noise emission level	

Safety standard compatibility:	UL508, EN61000-4-2 (ESD),	
Vibration (non-operating):	Swept Sine (EN60068-2-6): 2G, 18-500 Hz, 3 primary axes. Random: 20-500Hz	
Protection class:	IP20 protection class (minimum)	
Output protection fuse:	No output protection fuse. Short circuit protection provided with electronic hardware design	
Terminal Protection:	Terminals are inset within the plastic enclosure. Protected against accidental touch.	
<u>Standards Agency Compliance:</u>		
All models:	CE compliant (LVD, EMC and RoHS directives, no certification) UL compliant (no certification)	
<u>Additional functions:</u>		
Alarm Output:	Relay contact, coil activated during normal operation (energized for power good signal)	
	N/O contact relay	Relay contact, potential-free N/O contact (energized and closed under normal operating conditions) Contact capacity: 230Vac/1A max.
Alarm Functions:	AC/DC failure:	Relay de-energized, if: 1. AC power input is lost 2. If charger DC output is lost
	Low DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC}<88\%V_{FLOAT}$ 2. Alarm RESET if $V_{DC}>92\%V_{FLOAT}$
	High DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC}>107\%V_{FLOAT}$ 2. Alarm RESET if $V_{DC}<105\%V_{FLOAT}$
	Overvoltage:	Relay de-energized, if: 1. $V_{DC}>16V_{DC}$ (on 12VDC charger) 2. $V_{DC}>30.5V_{DC}$ (on 24VDC charger)
	Battery Disconnect:	Relay de-energized, if: Battery load is disconnected (alarm time delay 10 to 20 minutes)
Communication:	USB com port	Type-B USB socket, non-isolated, for device configuration purpose only.
Temperature measurement:	Pt-100 RTD sensor	Range: -40°C to +100°C for battery terminal voltage compensation (sensor not supplied)
Boost Enable/Disable:	Automatic BOOST function	Integrated "Automatic Boost" function, with an additional external BOOST control input (user selectable)
Charger stacking:	Serial connection:	Not allowed
	Parallel connection:	Multiple units of same model can be connected in parallel to increase total output current value and/or to introduce redundancy. Current sharing is based on V-I characteristics of each individual charger unit in the stack.

Label:	Printed, compliant to UL standards	Stamp printed on front panel cover	Model number Serial number, terminal definition Input limits Warnings Compliance, LED definition Alarm output indication.
Input Surge Protection:	Live – Earth	Capacitive decoupling	Common mode noise filter
	Neutral – Earth		
	Live-Neutral	MOV protected	
Earth Leakage Current:	Live-Neutral to Earth:	<1mA	50Hz – 60Hz ±10%
Plastic enclosure:	Nylon 66 plastic material with GF reinforcement	Black / 25%GF	No additional colouring
		UL94 (1,5mm HB)	850°C glow wire test compliant
	Mounting	DIN rail mount	Screw mounting with wall-mount plate
		Screw mount	
Mechanical dimensions:	L x W x H	To be determined during design phase	

REVISED TECHNICAL SPECIFICATIONS FOR 75W MODEL:

- USB non-isolated communication port added with type-2 socket
- Temperature compensation RTD sensor port added
- External BOOST control terminal added
- Foldback current limiting changed to full rated current into short circuit

MODEL-B: 12V/10A & 24V/5A (150W Package)

This is the model with dual-battery voltage selection. The topology is based on flyback conversion and targeted efficiency will be around 86% at 220Vac input and more than 50% output load.

Specifications	Description		
<u>Input Specifications:</u>			
AC Supply Input Voltage range:	Universal AC power input (withstanding to 300Vac)	88.0Vac min.	265Vac max.
		@ Full load conditions	
Operating Input Frequency range:	50Hz / 60Hz	45Hz min.	70Hz max.
DC Input voltage range:	DC volts	120V min.	400V max.
		@ full load conditions	
Continuous Input current:	@ Full load	90Vac input	<3.23Aac
		220Vac input	<1.22Aac

Efficiency:	>83% worst case condition		
Rated input power:	<178W @ full load and worst-case efficiency conditions		
Input Power Factor:	>0.65 Capacitive (no PFC compensation)		
Input fuse protection:	PCB mounted, non-replaceable fuse (no user serviceable parts)		
Input terminal type:	12V/10A 24V/5A	Mains	GSTB 2.5 / 3 Three pole Phoenix screw type terminal set
		Battery	MSTB 2.5 / 6 six pole Phoenix screw type terminal set
<u>Output Specifications:</u>			
Output voltage (V _{Float}):	12V selection (user configurable)	Factory set at 13.8Vdc Boost mode: 14.7Vdc	
	24V selection (user configurable)	Factory set at 27.6Vdc Boost mode: 29.4Vdc	
Output current range:	12V selection	10A	
	24V selection	5A	
Short-circuit current:	Full rated current into short circuit, continuous operation		
Output ripple voltage:	<1.0% of nominal output voltage over 10Hz to 100KHz		
Current limit:	Automatic current limitation at rated o/p current value, based on selected battery voltage setting.		
Boost mode control:	Built-in automatic boost function control, BOOST charge current limit to be decided during design phase.		
Output noise level:	<1.5V pk-pk (30KHz to 300KHz)		
Line regulation:	<1% over specified input voltage range at constant output load (not exceeding rated load)		
Load regulation:	<1% at constant input voltage, from no load to full load on the output, measured at the output terminals, no current limiting conditions		
Output Voltage ripple:	<160mV pk-pk at full load		
Battery voltage selection:	Battery voltage selection is user configurable from USB communication port		
Reverse polarity protection:	12V/10A 24V/5A	"Automatic Resettable" electronic reverse polarity protection. No external components (fuse) required.	
Front panel indicators:	Multicolor LED indicator for status and alarm conditions: • Bulk, Absorption and Float modes: steady green • Boost mode: blinking green • Failure: steady red Battery voltage selection: • Green LED for 12V and 24V selected battery voltage indication		
Auxiliary output:	Alarm relay output		N/O contact, 250Vac@1A AC3 class
<u>Environmental conditions:</u>			
Operating temperature range:	-40°C to +70°C with power derating		
Output Power derating:	Output power is linearly de-rated starting at 55°C down to 50% of its rated capacity at 70°C. (7pprox... -3.5%/°C power derating above 55°C)		

Storage temperature and humidity:	-40°C to +85°C, (0%rh to 95%rh, non-condensing)	
Cooling type:	Convection cooled (vertical mounted and no physical obstacles are allowed within 30mm from each side and 50mm between top and bottom sides)	
Operating humidity range:	0%Rh to 95%Rh, non-condensing and AC supply connected. Derating may activate below 30%RH depending on operating conditions	
Altitude:	2000mt at full rated load. De-rate linearly from full power to 50% power from 2000mt to 3000mt over the specified working temperature range.	
Dielectric withstanding voltage:	Mains input – DC output, Boost input & Alarm o/p (with voltage input terminals shorted) Reinforced-insulation between primary and secondary circuits.	4.0 KV _{DC} @ 70% RH
	Mains input – Earth (voltage input terminals shorted)	2,5 KV _{DC} @ 70% RH
	DC Output – Earth (with output terminals shorted)	0,5 KV _{AC} @ 70% RH
	Alarm o/p - DC out & Boost terminals	2,5 KV _{DC} @ 70% RH
ESD protection:	8kV free air discharge onto plastic front cover and metal case (metal case connected to Earth)	
EMI noise immunity:	IEC/EN 61000-6-2,	
Conducted and radiated EMC:	IEC/EN61000-6-4, IEC/EN61000-4-4, IEC/EN61000-4-5, IEC/EN61000-4-6, IEC/EN61000-4-11, IEC/EN61000-3-2, Light industry noise emission level	
Safety standard compatibility:	UL508, EN61000-4-2 (ESD),	
Vibration (non-operating):	Swept Sine (EN60068-2-6): 2G, 18-500 Hz, 3 primary axes.	
Environmental Protection class:	IP20 protection class (minimum)	
Output protection fuse:	No output protection fuse. Short circuit protection with electronic hardware	
Terminal Protection:	Terminals are inset within the plastic enclosure. Protected against accidental touch.	
Standards Agency Compliancy:		
All models:	CE compliant (LVD, EMC and RoHS directives, no certification) UL safety standard compliant (no certification)	
Additional functions:		
Alarm Output:	Relay contact, active during normal operation (energized for power good signal)	
	Relay fitted as standard	Relay contact, potential-free N/O contact (Energized under normal operating conditions) Contact capacity: 230Vac/1A max.
Alarm Functions:	AC/DC failure:	Relay de-energized, if: 1. AC power input is lost 2. If charger DC output is lost
	Low DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC}<88\%V_{FLOAT}$ 2. Alarm RESET if $V_{DC}>92\%V_{FLOAT}$

	High DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC}>107\%V_{FLOAT}$ 2. Alarm RESET if $V_{DC}<105\%V_{FLOAT}$	
	Overvoltage:	Relay de-energized, if: 1. $V_{DC}>16V_{DC}$ (on 12V charger) 2. $V_{DC}>30.5V_{DC}$ (on 24V charger)	
	Battery Disconnect:	Relay de-energized, if: Battery load is disconnected (alarm time delay, approx. 10 to 20 minutes)	
Boost Enable/Disable:	Automatic BOOST function	Integrated “Automatic Boost” function, with an additional external BOOST control input (user selectable)	
Communication:	USB com port	Type-B USB socket, non-isolated, for device configuration purpose only.	
Temperature measurement:	Pt-100 RTD sensor	Range: -40°C to +100°C for battery terminal voltage compensation (sensor not supplied)	
Charger stacking:	Serial connection:	Not allowed	
	Parallel connection:	Multiple similar models can be connected in parallel to increase total output current value and/or to introduce redundancy. Current sharing is based on V-I characteristics of individual charger in the stack.	
Label:	Stamp printed, compliant to UL standards	Printed on front cover	Model number Serial number Terminal definition Input limits Warnings Compliance, LED definition Alarm output indication,
Input Surge Protection:	Live – Earth	Capacitive decoupling	Common mode noise filter in the input circuit
	Neutral – Earth		
	Live-Neutral	MOV protected	
Earth Leakage Current:	Live-Neutral to Earth:	<1mA	50Hz – 60Hz ±5%
Plastic enclosure:	Nylon 66 with GF reinforcement filling	Black / 25%GF	no additional colouring
		UL94 (1,5mm HB)	850°C glow wire test compliant
	Mounting	DIN rail mount	Screw mounting with wall-mount plate.
		Screw mount	
Mechanical dimensions:	L x W x H	Mechanical dimensions will be determined during design phase	

REVISED TECHNICAL SPECIFICATIONS FOR 150W MODEL:

- USB non-isolated communication port added with type-2 socket
- Temperature compensation RTD sensor port added
- External BOOST control terminal added
- Foldback current limiting changed to full rated current into short circuit

MODEL-C: 24V/10A (300W package)

This package is a 24V model and will be based on forward conversion topology. Input stage will be designed with PFC circuit and targeted efficiency will be above 88% at 220Vac input voltage and above 50% output load.

Specification	Description		
<u>Input Specifications:</u>			
AC Supply Input Voltage range:	Universal input (withstand to 300Vac)	88.0Vac min.	265Vac
		@ Full load condition	
Operating Input Frequency range:	50Hz / 60Hz	45Hz min.	70Hz max.
DC operating input voltage range:	DC Volts	120VDC min @ full load conditions	400VDC max. @ full load conditions
Continuous input current at full load condition:	300W package (Cosφ >0.97 with PFC correction circuit)	90Vac input	<4.40Aac
		220Vac input	<1.85Aac
Efficiency:	>88% worst-case condition		
Input Power Factor:	>0.97 with “Power Factor Corrected” input circuit design		
Input fuse protection:	PCB mounted, replaceable glass fuse protection on the input circuit		
Input terminal type:	24Vdc model only	Mains	GSTB 2.5 / 3 (12Adc) Three pole screw type Phoenix terminal set
		Battery	MSTB 2.5 / 6 (12Adc) six pole screw type Phoenix terminal set
<u>Output Specifications:</u>			
Output voltage:	Float voltage setting	Factory setting: 27.6Vdc	
	Boost voltage setting	Factory setting: 29.0Vdc	
Output current range:	24vdc	0 – 10Adc load dependent	
Output terminal types:	PCB mounted	16Adc	Phoenix Screw type terminal set
Short-circuit current:	Rated output current (10Adc) into short circuit, continuous operation		
Output ripple voltage:	<1.0% of nominal output voltage over 10Hz to 100KHz		
Current limit:	100% rated current during BULK and ABSORPTION charging modes		
Boost mode control:	Built-in automatic boost function, with one EXTERNAL Boost control input. Boost charging current derating will be decided during design phase.		
Output noise level:	<1.5V pk-pk (30KHz to 300KHz)		
Line regulation:	<1.0% over 100Vac to 300Vac between -20°C to 55°C at 50% constant load condition		
Load regulation:	<1.0% at constant input voltage between 115Vac and 265Vac, from 10% load to 90% load on the output, measured at the output terminals over -20°C and 55°C operating temperature		
Output Voltage ripple:	<150mV pk-pk at full load conditions		

Reverse polarity protection:	With 24V battery connected to output terminals	“Automatic Resettable” electronic reverse polarity protection. No external components (fuse) required.
Front panel indicators:	Multicolor LED indicator for status and alarm conditions: • Bulk, Absorption and Float mode charging: steady green • Boost mode: blinking green • Failure: steady red	
Auxiliary output:	Alarm relay output	N/O contact, 250Vac@2A AC3 class
		N/C contact, 250Vac@1A AC3 class
Environmental conditions:		
Operating temperature range:	-40°C to +70°C ambient temperature. Starting at 55°C ambient, output power will be linearly derated down to 50% of rated output power at 70°C	
Output Power derating:	Output power is linearly de-rated starting at 55°C down to 50% of its rated capacity at 70°C. Output will cut off at T _{AMB} > 70°C	
Storage temperature and humidity:	-40°C to +85°C, (10%rh to 95%rh, non-condensing)	
Cooling type:	Convection cooled (vertical mounted and no physical obstacles are allowed within 30mm from each side and 50mm above and below)	
Humidity:	0%Rh to 95%Rh, non-condensing and AC supply connected. (power derating may become active below 30%RH)	
Altitude:	Up to 2000mt at full rated load (input voltage >110Vac) De-rate linearly from full power to 50% power from 2000mt to 3000mt over the specified operating temperature range.	
Dielectric withstanding voltage:	Mains input – DC output, Boost input & Alarm o/p (with voltage input terminals shorted) Reinforced insulation between primary and secondary circuits.	4.0 KV _{DC} @ 70% RH
	Mains input – Earth (voltage input terminals shorted)	2,5 KV _{DC} @ 70% RH
	DC Output – Earth (with output terminals shorted)	0,5 KV _{AC} @ 70% RH
	Alarm o/p - DC out & Boost terminals	2,5 KV _{DC} @ 70% RH
ESD protection:	8kV free air discharge onto the metal case and plastic enclosure (metal case connected to EARTH)	
EMI noise immunity:	IEC/EN 61000-6-2,	
Conducted and radiated EMC:	IEC/EN61000-6-4, IEC/EN61000-4-4, IEC/EN61000-4-5, IEC/EN61000-4-6, IEC/EN61000-4-11, IEC/EN61000-3-2, Light industry noise emission level	
Safety standard compliance:	UL508, EN61000-4-2 (ESD),	
Vibration (non-operating):	Swept Sine (EN60068-2-6): 4G, 18-500 Hz, 3 primary axes. Random: 20-500Hz,	
Environmental Protection class:	IP20 protection class (minimum)	
Output protection fuse:	No output protection fuse. Short circuit protection with electronic hardware	
Terminal Protection:	Terminals are inset within the plastic enclosure. Protected against accidental touch.	

Standards Agency Compliance:			
All models:	CE compliant (LVD, EMC and RoHS directives, no certification) UL compliant (no certification)		
Additional functions:			
Alarm Output:	Active during normal operation (energized for power good signal)		
	Relay fitted as standard (SPDT relay)	Relay contact, potential-free N/O - N/C contact (energized under normal operating conditions) Contact capacity: 230Vac/1Aac max.	
Alarm Functions:	AC/DC failure:	Relay de-energized, if: 1. AC power input is lost 2. If charger DC output is lost	
	Low DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC}<88\%V_{FLOAT}$ 2. Alarm RESET if $V_{DC}>92\%V_{FLOAT}$	
	High DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC}>107\%V_{FLOAT}$ 2. Alarm RESET if $V_{DC}<105\%V_{FLOAT}$	
	Overvoltage:	Relay de-energized, if: 1. $V_{DC}>16V_{DC}$ (on 12V charger) 2. $V_{DC}>30.5V_{DC}$ (on 24V charger)	
	Battery Disconnect:	Relay de-energized, if: Battery load is disconnected (alarm time delay, approx. 10 to 20 minutes)	
Boost Enable/Disable:	Automatic BOOST function	Integrated "Automatic Boost" function with an additional terminal for external BOOST control	
Communication:	USB com port	Type-B USB socket, non-isolated, for device configuration purpose only.	
Temperature measurement:	Pt-100 RTD sensor	Range: -40°C to +100°C for battery terminal voltage compensation (sensor not supplied)	
Charger stacking:	Serial connection:	Not allowed	
	Parallel connection:	Multiple similar units can be connected in parallel to increase total output current value and/or to introduce redundancy. Current sharing is based on V-I characteristics of individual charger in the stack.	
Label:	Printed, compliant to UL standards	Stamp printed on enclosure front cover	Model number Serial number Terminal definition Input limits Warnings Compliance LED definition Alarm output
Input Surge Protection:	Live - Earth	Capacitive coupling	Common mode noise filter stage for conduction EMC
	Neutral - Earth		
	Live-Neutral	MOV protected	
Earth Leakage Current:	Live-Neutral to Earth:	<1mA	50Hz – 60Hz ±5%
Plastic enclosure:	Nylon-6 25%GF	Original Black granules	No coloring
		UL94 (1.5mm HB)	850°C glow wire test compliant

	Mounting	DIN rail mount	Standard feature
		Screw mount with mounting plate	
Mechanical dimensions:	L x W x H	To be determined during design phase	

REVISED TECHNICAL SPECIFICATIONS FOR 300W MODEL:

- PFC input circuit
- USB non-isolated communication port added with type-2 socket
- Temperature compensation RTD sensor port added
- External BOOST control terminal added
- Foldback current limiting changed to full rated current into short circuit

MODEL-D: 24V/20A (600W package)

This model will be designed with communication capability and topology will be based on LLC resonant converter technology to achieve high efficiency

Specification	Description		
<u>Input Specifications:</u>			
AC Supply Input Voltage range:	Universal input (withstanding 300Vac)	88.0Vac min.	265Vac max.
		Full load	
Operating Input Frequency range:	45Hz to 70Hz. (All models can operate at full capacity with DC input voltage feed, within their specified input voltage ranges)		
DC Input voltage range:	DC Volts	120VDC min. @ full load conditions	400VDC max. @ full load conditions
Continuous Input current at full load condition:	600W package (Cosφ > 0.97)	90Vac input	<7.8Aac
		220Vac input	<3.0Aac
Efficiency:	>90% worst case condition		
Input Power Factor:	>0.97 PFC corrected input stage		
Input fuse protection:	PCB mounted, replaceable glass fuse, not front panel accessible.		

Input / Output terminal types:	24Vdc model only	Mains	GSTB 2.5 / 3 (12Adc) Three pole screw type Phoenix terminal set
		Battery	40A stud type battery terminals
		Control and signal	MSTB2.5/6 screw type Phoenix terminals. The number of terminals may vary depending on the required number of external control pins.
Output Specifications:			
Output voltage range:	24Vdc nominal	Factory set at 27.6Vdc	For BULK, ABSORPTION and FLOAT mode operation
		Boost: 29.0Vdc	Boos mode only (gassing limit)
Output current range:	DC amps	0 to 20A depending on output load	
Output terminal types:	24VDC	40.0A (contact capacity)	Stud type screw terminal set
Short-circuit current:	Output current cut off below 60% of rated open circuit output voltage level. Full rated current can be delivered, provided that; loaded output voltage is >0.6V _{OUT-nominal} .		
Output ripple voltage:	<1.5% of nominal output voltage over 10Hz to 300KHz		
Current limit:	100% rated current in Bulk / Absorb charging modes 50% rated current in Boost mode, governed by rated maximum power and operating conditions,		
Boost mode control:	Intelligent BOOST charge mode based on integrated BMS software algorithm. There is an additional terminal for EXTERNAL Boost function control		
Charger stacking:	Series connection	Not allowed	
	Parallel connection	Multiple same model chargers can be connected in parallel to increase output charge current and/or to introduce redundancy. Current sharing of each unit depends on the V-I characteristics of individual charger unit. Intelligent current sharing possible when connected with RBM or EDU accessories.	
Output noise level:	<1.5V pk-pk (30KHz to 300KHz)		
Line regulation:	<1% over specified input voltage range (90Vac to 265Vac) at constant output load of 50%		
Load regulation:	<1.5% at constant input voltage of 220Vac, from 10% load to 90% load on the output, measured across output terminals of the charger unit,		
Reverse polarity protection:	All models		"Automatic Resettable" electronic reverse polarity protection. No external components (fuse) required.
Front panel LED indicators:	Multicolor LED indicator for status and alarm conditions: • Bulk, Absorption and Float mode: steady green • Boost mode: blinking green • Failure: steady red • Communication port status indicator • Auxiliary i/o status indicators (high battery temperature)		
Alarm output:	Alarm relay output (SPDT)	N/O contact, 250Vac@1A AC3 class	
		N/C contact, 250Vac@1A AC3 class	
AUXILIARY I/O:	Digital I/O	2 digital inputs and 1 digital output (solid state protected output) available for external device integration. All AUX i/o connections are non-isolated and rated voltage is 24Vdc.	

Environmental conditions:		
Operating temperature range:	-40°C to +70°C, between 200Vac to 440Vac input voltage range with power derating active	
Operating humidity range:	0%RH to 95%RH (power derating may activate below 30%RH based on the operating conditions), AC power is connected,	
Output Power derating:	Output power is linearly de-rated starting at 55°C down to 50% of its rated capacity at 70°C. Power derating factor is approx. -3.5%/°C from 55°C to 70°C ambient temp.	
Storage temperature and humidity:	-40°C to +85°C, (0%rh to 95%rh, non-condensing)	
Cooling type:	Convection cooled (vertical mounted and no physical obstacles are allowed within 50mm from each side and 100mm above and below.	
Altitude:	2000mt at full rated load. De-rate linearly from full power to 50% power from 2000mt to 3000mt over the specified working temperature and humidity range.	
Dielectric withstanding voltage:	Mains input – DC output, Boost input & Alarm o/p (with voltage input terminals shorted) Reinforced-insulation between primary and secondary circuits.	4.0 KV _{DC} @ 70% RH
	Mains input – Earth (voltage input terminals shorted)	2,5 KV _{DC} @ 70% RH
	DC Output – Earth (with output terminals shorted)	0,5 KV _{AC} @ 70% RH
	Alarm o/p - DC out & Boost terminals	2,5 KV _{DC} @ 70% RH
ESD protection:	8kV free air discharge onto plastic enclosure and metal parts (with unit connected to EARTH)	
EMI noise immunity:	IEC/EN 61000-6-2,	
Conducted and radiated EMC:	IEC/EN61000-6-4, IEC/EN61000-4-4, IEC/EN61000-4-5, IEC/EN61000-4-6, IEC/EN61000-4-11, IEC/EN61000-3-2, Light industry noise emission level	
Safety standard compliance:	UL508, EN61000-4-2 (ESD),	
Vibration (non-operating):	Swept Sine (EN60068-2-6): 4G, 18-500 Hz, 3 primary axes. Random: 20-500Hz,	
Environmental Protection class:	IP20 protection class (minimum)	
Output protection fuse:	No output protection fuse. Short circuit current limiting with electronic hardware	
Terminal Protection:	Terminals are inset within the plastic enclosure. Protected against accidental touch.	
Standards Agency Compliance:		
All models:	CE compliant (LVD, EMC and RoHS directives, no certification) UL compliant (no certification)	
Additional functions:		
Alarm Output:	Active during normal operation (energized for power good signal)	
	SPDT Relay fitted as standard on all models	Relay contact, potential-free N/O - N/C contact (energized under normal operating conditions) Contact capacity: 230Vac/1A AC3 class
Alarm Functions:	AC/DC failure:	Relay de-energized, if: 1. AC power input is lost 2. If charger DC output is lost

	Low DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC} < 88\% V_{FLOAT}$ 2. Alarm RESET if $V_{DC} > 92\% V_{FLOAT}$	
	High DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC} > 107\% V_{FLOAT}$ 2. Alarm RESET if $V_{DC} < 105\% V_{FLOAT}$	
	Overvoltage:	Relay de-energized, if: 1. $V_{DC} > 30.5 V_{DC}$ (on 24V charger)	
	Battery Disconnect:	Relay de-energized, if: Battery load is disconnected (alarm time delay, approx. 10 to 20 minutes)	
	High Battery temperature:	Relay de-energized if: $T_{BATTERY} > SET\ TEMP.\ POINT$	
	S/W Alarm warnings:	When connected with COM port, additional alarm and status indication data will be available with PC Tool Software. BMS measured and calculated data will be available from the COM port only.	
Boost Enable/Disable:	Automatic BOOST control	Integrated "Automatic Boost" control function plus an additional EXTERNAL Boost control terminal	
Label:	Printed, compliant to UL standards	Stamp printed on front cover	Model number Serial number Terminal definition Input limits Warnings Compliance LED definition Alarm output indication
Input Surge Protection:	Live - Earth	Capacitive coupling	Common mode noise filter and PFC circuit for conducted EMC compatibility
	Neutral - Earth		
	Live-Neutral	MOV protected	
Earth Leakage Current:	Live-Neutral to Earth:	<1mA	50Hz – 60Hz ±5%
<u>Communications:</u>			
USB (type2):	For device configuration only (no galvanic isolation, for device configuration only) Type-B USB socket on the PCB.		
CAN BUS (CAN open):	For connection to peripheral devices and networks. (Galvanically isolated communications and power input)		
Battery temperature measuring input:	Pt-100 RTD temperature sensor	Configurable with PC Tool S/W for battery temperature measurement. (sensor not supplied)	
Enclosure:	Nylon-6 with GF reinforcement and aluminum heatsink plate	Black	Natural granules, no color granule additives
		UL94 (1,5mm HB)	850°C glow wire test compliant
	Mounting	Screw mount with mounting plate	Standard feature

REVISED TECHNICAL SPECIFICATIONS FOR 600W MODEL:

- External BOOST control terminal added

MODEL-E: 24V/40A (1200W package)

These chargers will be designed with 3-phase input stage and PFC correction circuit. They will be panel mounted (not suitable for DIN rail mounting).

Specification	Description		
<u>Input Specifications:</u>			
AC Supply Input Voltage range:	3-phase input (no NEUTRAL connection)	200Vac min.	440Vac max.
		Full load conditions	
Operating Input Frequency range:	45Hz to 70Hz.		
DC Input voltage range:	DC Volts	300VDC min. and 550VDC max. Input rectifier diode current rating will be selected according to DC input current values connected across two phases.	
Continuous Input current at full load:	1200W package (Cosφ > 0.97)	3 x 200Vac input	<3.6Aac
Efficiency:	>93% worst case condition		
Input Power Factor:	>0.97 with PFC corrected input stage		
Input fuse protection:	PCB mounted, replaceable fuse (not suitable for user servicing)		
Input / Output terminal types:	24Vdc model only	Mains	1000V rated screw type terminal strip
		Battery	100A stud type battery terminals
		Control and signal	Phoenix MSTB2.5 screw type terminals (required number of terminals)
<u>Output Specifications:</u>			
Output voltage range:	24Vdc nominal	Factory set at 27.6Vdc Boost mode: 29.0Vdc	Configurable via PC Tool software

Output current range:	DC Amps	0-40A load dependent
Short-circuit current:	Output current cut off below 60% of rated open circuit output voltage level. Full rated current can be delivered, if; loaded output voltage is $>0.6V_{OUT-nominal}$.	
Output ripple voltage:	$<1.5\%$ of nominal output voltage over 10Hz to 300KHz	
Current limit:	100% rated current in Bulk / Absorption and Float charging phases. Output current will cut-off if battery terminal voltage is $<60\%$ of nominal battery voltage level.	
Boost mode control:	Intelligent BOOST charge mode based on integrated BMS software algorithm. Boost mode can also be configured and controlled via communication ports.	
Output noise level:	$<1.2V$ pk-pk (30KHz to 300KHz) at 50% load level.	
Line regulation:	$<1\%$ over specified input voltage range at 50% output load condition.	
Load regulation:	$<1.5\%$ at nominal input voltage, from 10% load to 90% load on the output, measured at the output terminals	
Output Voltage ripple:	$<300mV$ pk-pk at full load (not in current limit conditions)	
Reverse polarity protection:	Solid state protection technology	"Automatic Resettable" electronic reverse polarity protection. No external components (fuse) required.
Front panel indicators:	Multicolor LED indicator for status and alarm conditions: • Float mode: steady green • Boost mode: steady blinking green • Failure: steady red • Communication status indicators • i/o status indicators (high battery temperature)	
Alarm output:	SPDT Relay	N/O contact, 250Vac@2A AC3 class
		N/C contact, 250Vac@1A AC3 class
Auxiliary input / output:	24Vdc Digital input and output ports	2 digital inputs and 1 solid-state digital output, non-isolated and rated at 24Vdc. Configurable via PC tool software.
<u>Environmental conditions:</u>		
Operating temperature range:	$-40^{\circ}C$ to $+70^{\circ}C$ ambient temperature.	
Output Power derating:	Output power is linearly de-rated starting at $55^{\circ}C$ down to 50% of its rated output capacity at $70^{\circ}C$. Output will cut off at around $T_{AMB} > 70^{\circ}C$	
Storage temperature range:	$-40^{\circ}C$ to $+85^{\circ}C$, (0%rh to 95%rh, non-condensing)	
Cooling type:	Convection cooled (vertical mounted and no physical obstacles are allowed within 50mm from each side and 100mm above and below)	
Operating Humidity range:	0%Rh to 95%Rh, non-condensing and AC supply connected. Power derating may activate if humidity is $<30\%RH$ under certain operating conditions	
Altitude:	2000mt at full rated load. De-rate linearly from full power to 50% rated power from 2000mt to 3000mt over the specified working temperature range.	
Dielectric withstanding voltage:	Mains input – DC output, Boost input & Alarm o/p (with voltage input terminals shorted) Reinforced insulation between primary and secondary circuits.	4.0 KV _{DC} @ 70% RH

	Mains input – Earth (voltage input terminals shorted)	2,5 KV _{DC} @ 70% RH
	DC Output – Earth (with output terminals shorted)	0,5 KV _{AC} @ 70% RH
	Alarm o/p - DC out & Boost terminals	2,5 KV _{DC} @ 70% RH
ESD protection:	8kV free air discharge onto the metal case (metal case is internally connected to EARTH)	
EMI noise immunity:	IEC/EN 61000-6-2,	
Conducted and radiated EMC:	IEC/EN61000-6-4, IEC/EN61000-4-4, IEC/EN61000-4-5, IEC/EN61000-4-6, IEC/EN61000-4-11, IEC/EN61000-3-2, Light industry noise emission level	
Safety standard compliance:	UL508, EN61000-4-2 (ESD),	
Vibration (non-operating):	Swept Sine (EN60068-2-6): 4G, 18-500 Hz, 3 primary axes. Random: 20-500Hz,	
Environmental Protection class:	IP20 protection class (minimum)	
Output protection fuse:	No output protection fuse. Short circuit protection with electronic hardware	
Terminal Protection:	Terminals are inset within the plastic enclosure. Protected against accidental touch.	
<u>Standards Agency Compliance:</u>		
All models:	CE compliant (LVD, EMC and RoHS directives, no certification) UL compliant (no certification)	
<u>Additional functions:</u>		
Alarm Output:	Active during normal operation (energized for power good signal)	
	SPDT Relay fitted as standard on all models	Relay contact, potential-free N/O - N/C contact (energized under normal operating conditions) Contact capacity: 230Vac/1A max.
Alarm Functions:	AC/DC failure:	Relay de-energized, if: 1. AC power input is lost 2. If charger DC output is lost
	Low DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC}<88\%V_{FLOAT}$ 2. Alarm RESET if $V_{DC}>92\%V_{FLOAT}$
	High DC output:	Relay de-energized, if: 1. Charger DC output voltage $V_{DC}>107\%V_{FLOAT}$ 2. Alarm RESET if $V_{DC}<105\%V_{FLOAT}$
	Overvoltage:	Relay de-energized, if: 1. $V_{DC}>30.5V_{DC}$ (on 24V charger)
	Battery Disconnect:	Relay de-energized, if: Battery load is disconnected (alarm time delay, approx. 10 minutes)
	High Battery temperature:	Relay de-energized if: $T_{BATTERY} > SET TEMP. POINT$

	S/W Alarm warnings:	When connected with COM port, additional alarm and status indication data will be available with PC Tool Software. BMS measured and calculated data will be available from the COM port only.	
Boost Enable/Disable:	All models	Integrated "Automatic Boost" control function. Can be also controlled via com-port with PC-Tool software. There is an additional terminal for EXTERNAL Boost control.	
Label:	Printed, compliant to UL standards	Stamp printed on front cover	Model number Serial number Terminal definition Input limits Warnings Compliance LED definition Alarm output and status indications
Input Surge Protection:	Live - Earth	Capacitive coupling	Common mode noise filter
	Neutral - Earth		
	Live-Neutral	MOV protected	
Earth Leakage Current:	Live-Neutral to Earth:	<1mA	50Hz – 60Hz ±5%
<u>Communications:</u>			
USB (type2):	For device configuration only (no galvanic isolation, for device configuration only) Type-B USB socket on the PCB.		
CAN BUS (CAN Open):	For connection to peripheral devices and networks. (Galvanically isolated communications and power input)		
Temperature measuring input:	Pt-100 RTD sensor	Configurable with PC Tool S/W for battery temperature measurement. (sensor not supplied)	
Power limit selection:	User configurable (600W / 1200W)	Output power can be limited to 600W (with current limit parameter selection) for applications with 3-phase input configuration.	
Enclosure:	Nylon-6 with GF reinforcement and aluminum metal heatsink	Black	Natural colored granule, no additional coloring
		UL94 (1,5mm HB)	850°C glow wire test compliant
	Mounting	Screw mount only on panel surface	Standard feature

REVISED TECHNICAL SPECIFICATIONS FOR 1200W MODEL:

- External BOOST control terminal added

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